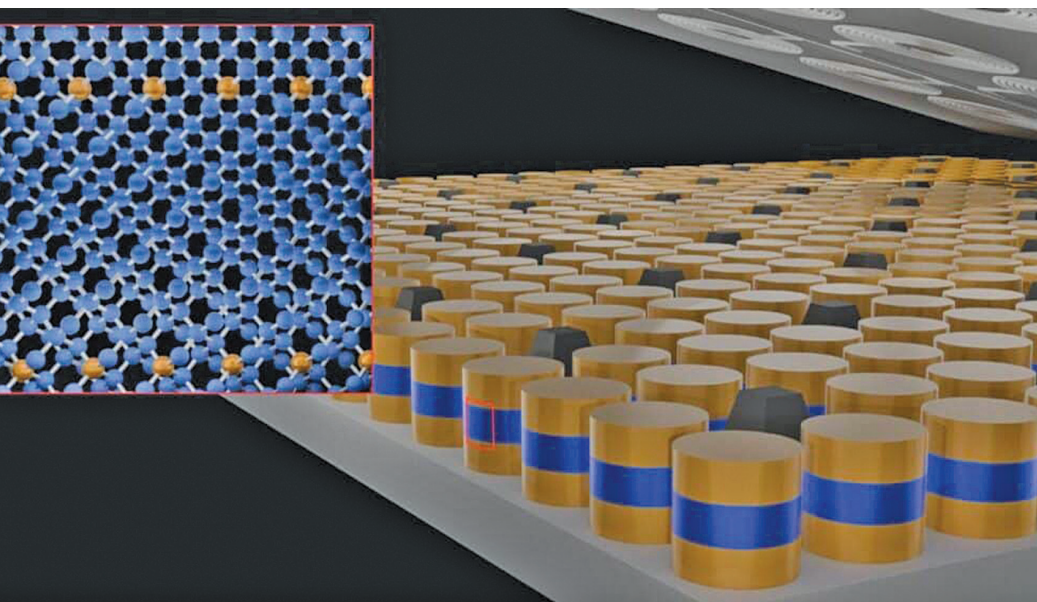




RESEARCH

& INNOVATION UPDATES

Breakthrough bridges semiconductor and superconducting technologies



Josephson junction structures—quantum devices made of two superconductors and a thin non-superconducting barrier—using different forms of germanium (Ge): super-Ge (in gold), semiconducting Ge (in blue), and super-Ge on wafer-level scale. Millions of Josephson junction pixels (10 micrometre square) can be created with this new material stack on wafer scale. Inset shows crystalline form of Super-Ge on the same matrix of semiconductor Ge (Credit: Patrick Strohbeen/NYU)

Researchers at New York University have transformed germanium, a common semiconductor, into a true superconductor, achieving zero electrical resistance at 3.5K. Using molecular beam epitaxy, the team infused gallium atoms deep into the lattice of germanium, surpassing normal doping limits while preserving the structure. The breakthrough bridges semiconductor and superconductor technologies, paving the way for ultra-efficient quantum circuits, cryogenic electronics, and scalable hybrid devices built on existing chip platforms.

New system for robots to navigate disaster zones

Researchers at MIT (Cambridge, Massachusetts) have unveiled a new mapping system that enables robots to navigate disaster zones more efficiently and accurately. Instead of processing limited images, it builds 3D submaps and stitches them into a full layout in real time, with no camera calibration or expert tuning needed. Using flexible mathematical transformations, it maintains under 5cm error margins. The system scales efficiently, supporting rescue missions, warehouse automation, and immersive XR applications.

The artificial intelligence-driven system incrementally creates and aligns smaller submaps of the scene, which it stitches together to reconstruct a full 3D map, like of an office cubicle, while estimating the robot's position in real time (Credit: Courtesy of the researchers)

